

Auto Ticket Classification using Flow Designer

Project Description:

Auto Ticket Classification using Flow Designer automates the way a school IT helpdesk sorts its support tickets. Students and teachers raise requests such as Wi-Fi issues, projector failures, password problems and slow computers. Instead of an IT agent reading every request and choosing the category by hand, a ServiceNow Flow Designer flow reads the Short Description at the time the ticket is created and fills in the Category and Subcategory automatically.

The solution uses a custom table (Incident Workflow) with an Auto Number, dependent Category and Subcategory choice fields, and a no-code flow built with Record Created trigger, If / Else If conditions, Update Record and Send Email actions. A confirmation email is sent to the caller as soon as the ticket is created. The whole configuration is captured in a Project Update Set so it can be exported as XML and moved to another instance.

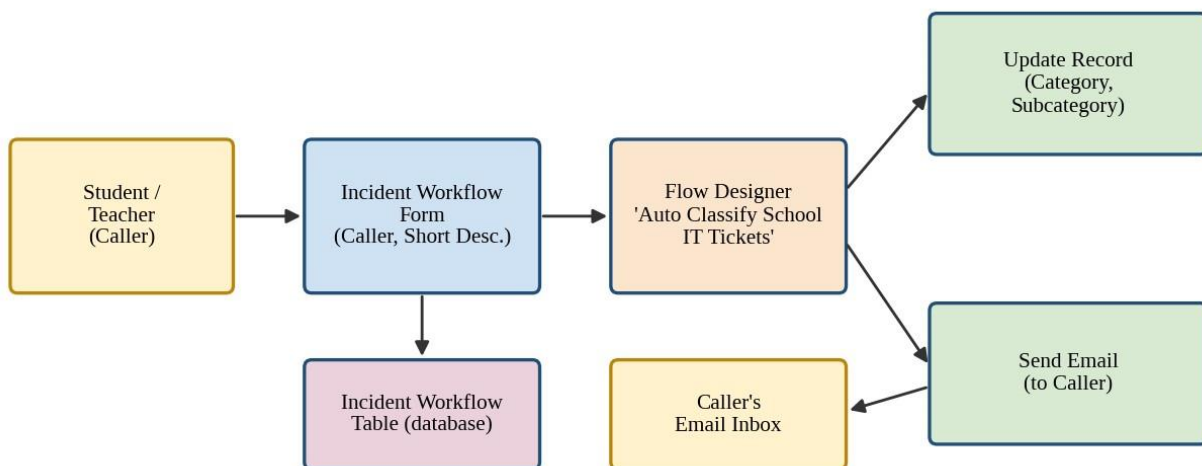
Scenarios

Scenario 1: A student enters the caller name and the short description "WiFi not working in library". The flow detects the Wi-Fi keyword, sets Category = Network and Subcategory = Wi-Fi, and emails a confirmation to the caller.

Scenario 2: A teacher reports "Projector not turning on". The flow sets Category = Hardware and Subcategory = Projector without any manual selection, and the caller receives an email.

Scenario 3: A student writes "Forgot password, cannot login" and another user writes "Computer is very slow". The flow classifies them as Access / Forgot Password and Performance / Slow Computer, so the IT staff can route them straight away.

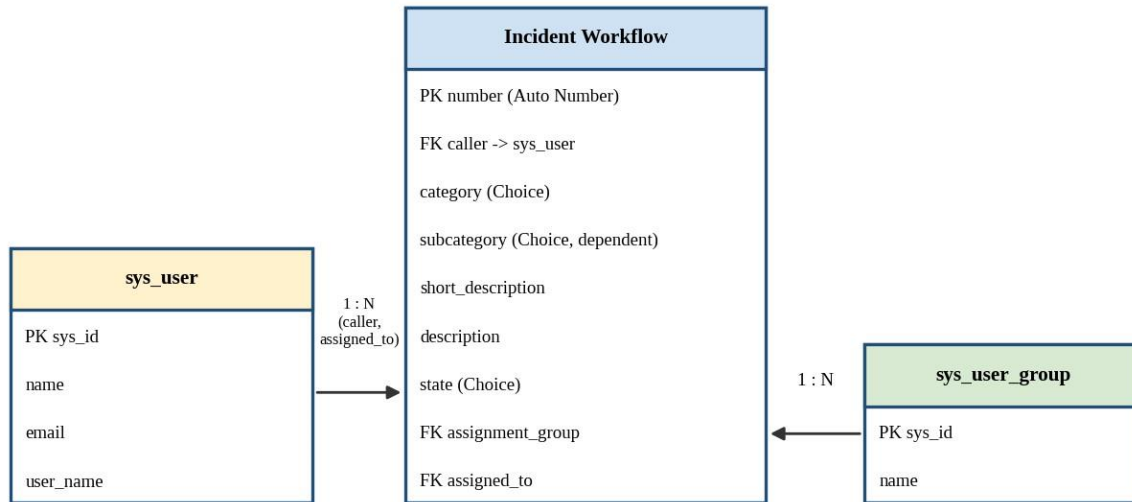
Technical Architecture:



Description: The solution follows a simple three-layer design on the ServiceNow platform. The caller uses the Incident Workflow form (presentation layer). When the record is created, the Flow Designer flow Auto Classify School IT Tickets starts (logic layer): it checks the Short Description, updates the Category and Subcategory and

sends an email to the caller. The ticket, the choice values and the references to users and groups are stored in the platform database (data layer).

ER Diagram:



Description: The custom table Incident Workflow stores one record per ticket. Its Caller and Assigned to fields reference the sys_user table, and its Assigned Group field references the sys_user_group table (one user or group can be linked to many tickets). Category, Subcategory and State are choice fields, and Subcategory depends on Category.

Pre-requisites:

- ServiceNow Personal Developer Instance with admin access: [ServiceNow Developer Site](#)
- Knowledge of Tables and Fields (Form Designer, Dictionary)
- Knowledge of Flow Designer (triggers, Flow Logic, actions)
- Basic understanding of Update Sets
- Knowledge of Notifications / Email in ServiceNow
- A modern web browser (Chrome / Edge)

Project Workflow:

Milestone 1: Requirement Analysis & Planning

- Activity 1.1: Understand the business use case and requirements.
- Activity 1.2: Create the Project Update Set.
- Activity 1.3: Navigation steps in picture view.

Milestone 2: Backend Development & Configuration

- Activity 2.1: Create the custom table to store ticket records.
- Activity 2.2: Create fields and configure data types.
- Activity 2.3: Implement the dependency between Category and Subcategory.

Milestone 3: Automation using Flow Designer & Email Notification

- Activity 3.1: Create the flow Auto Classify School IT Tickets.
- Activity 3.2: Configure the trigger.
- Activity 3.3 - 3.6: Wi-Fi, Projector, Password and Slow Computer classification.
- Activity 3.7: Set up the email action and activate the flow.

Milestone 4: Testing, Validation & Security

- Activity 4.1: Test scenario 1 (Wi-Fi).
- Activity 4.2: Check the email notification.
- Activity 4.3: Test scenario 2 (Projector).

Milestone 5: Deployment & Final Documentation

- Activity 5.1: Make the update set Complete and export it to XML.

Milestone 6: Conclusion

Milestone 1: Requirement Analysis & Planning

The school IT helpdesk receives many support requests every day from students and teachers about network connectivity, hardware failures, account access problems and system performance. At present IT staff manually review each ticket to identify the issue type, select the correct category and subcategory, and inform the caller that the ticket was created. This manual process is time-consuming, error-prone and not scalable as the number of requests grows.

Activity 1.1: Business Use Case

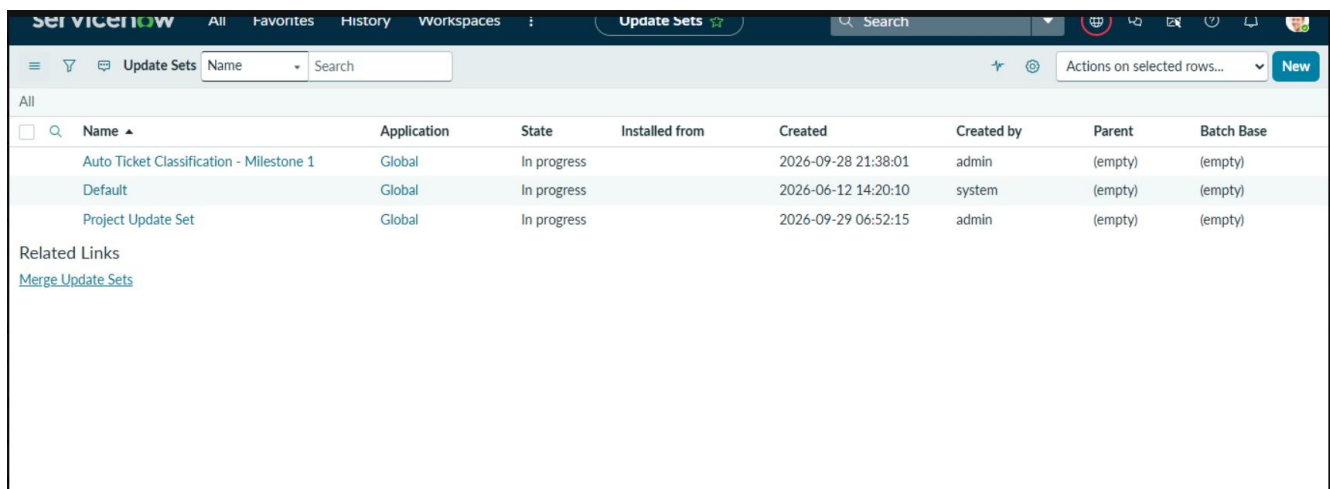
The system must:

1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

Activity 1.2: Creation of New Update Set Navigation

steps:

- Go to All menus and search for **Update set**.
- Under System Update Sets, select **Local Update Sets**.
- Click the **New** button on the top right corner of the page.
- Name: **Project Update Set**. State: **In progress**. Application: **Global**.
- Click **Save**.
- Under Related Links, click **Make This My Current Set**. The created update set now appears as the current set.

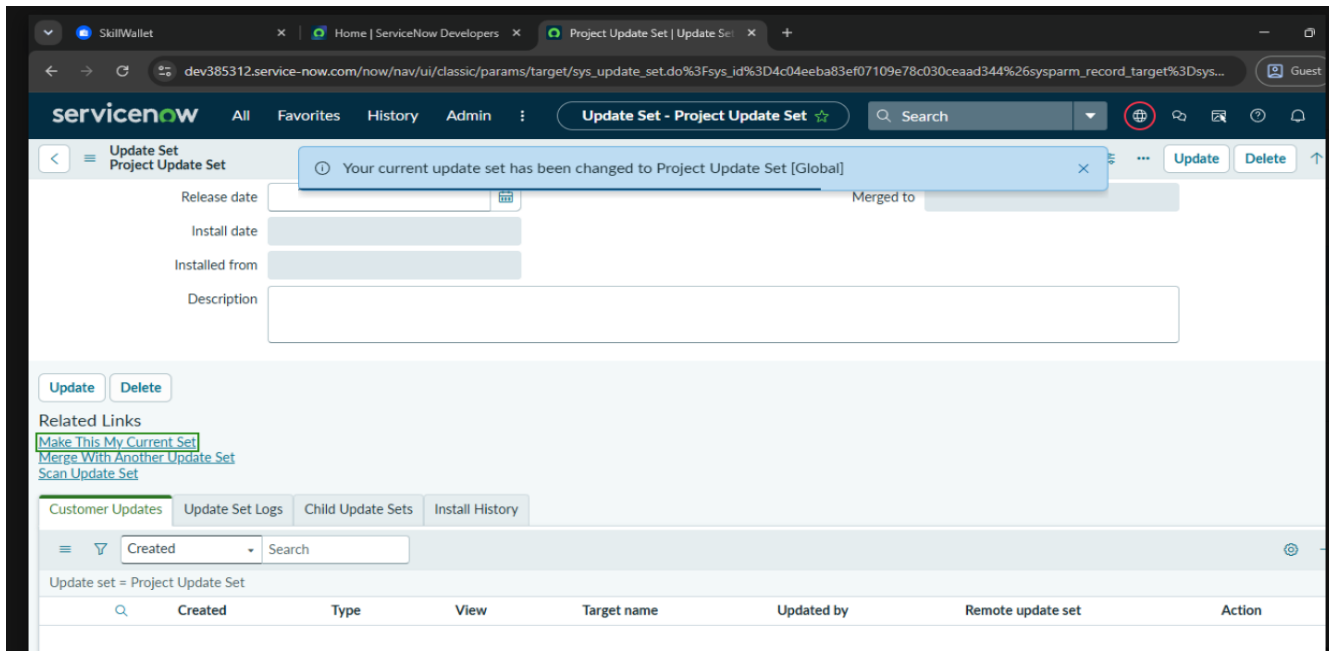


The screenshot shows the ServiceNow 'Update Sets' page. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and 'Update Sets'. A search bar is present. Below the navigation, there's a table of update sets. The table has columns: Name, Application, State, Installed from, Created, Created by, Parent, and Batch Base. Three update sets are listed: 'Auto Ticket Classification - Milestone 1', 'Default', and 'Project Update Set'. The 'Project Update Set' is highlighted. Below the table, there's a 'Related Links' section with a link to 'Merge Update Sets'.

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Auto Ticket Classification - Milestone 1	Global	In progress		2026-09-28 21:38:01	admin	(empty)	(empty)
Default	Global	In progress		2026-06-12 14:20:10	system	(empty)	(empty)
Project Update Set	Global	In progress		2026-09-29 06:52:15	admin	(empty)	(empty)

Related Links
[Merge Update Sets](#)

Activity 1.3: Navigation Steps in Picture



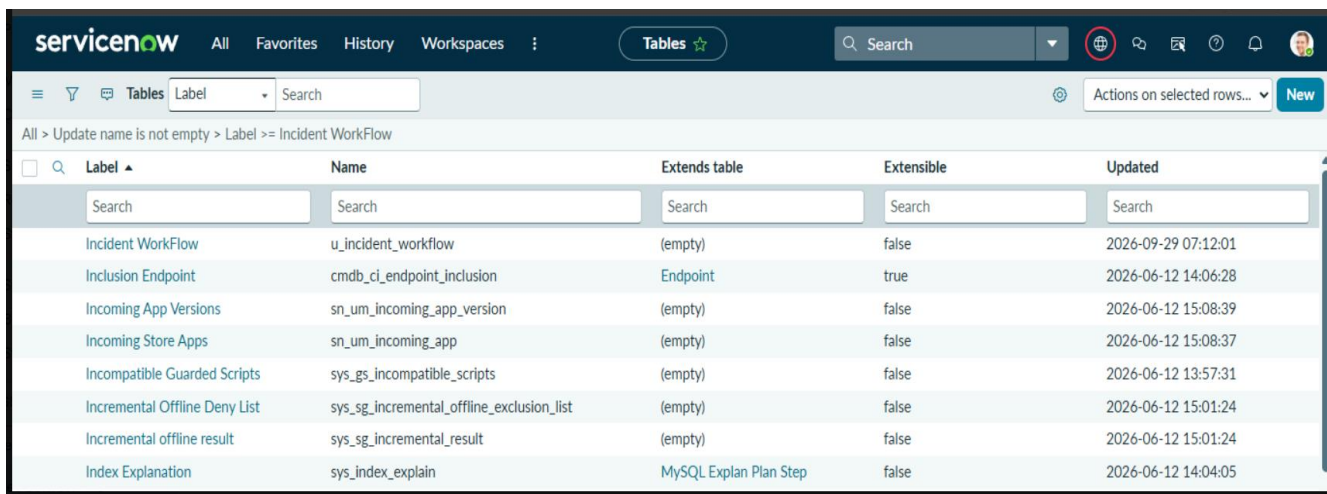
The screenshot shows the ServiceNow 'Update Set - Project Update Set' form. A blue notification bar at the top states: 'Your current update set has been changed to Project Update Set [Global]'. The form includes fields for 'Release date', 'Install date', 'Installed from', and 'Description'. Below these fields are 'Update' and 'Delete' buttons. A 'Related Links' section contains links: 'Make This My Current Set', 'Merge With Another Update Set', and 'Scan Update Set'. At the bottom, there are tabs for 'Customer Updates', 'Update Set Logs', 'Child Update Sets', and 'Install History'. A table below the tabs shows the 'Update set = Project Update Set' with columns: Created, Type, View, Target name, Updated by, Remote update set, and Action.

Milestone 2: Backend Development & Configuration

Activity 2.1: Creation of Custom Table to Store Ticket Records

Steps to create the custom table:

- Open All menus and search for **Tables**. • Under System Definition, click **Tables**.
- On the top right corner click the **New** button.
- Label: **Incident WorkFlow**. Uncheck **Create module**.
- Click **Save** the form.
- Open the **Controls** section and enable the check box **Auto Number**.
- Save the form again. Under Related Links click **Show Form**.
- Open the context menu, choose Configure and open the **Form Designer**. Create the fields as required.



The screenshot shows the ServiceNow 'Tables' list. The breadcrumb is 'All > Update name is not empty > Label >= Incident WorkFlow'. The table has columns: Label, Name, Extends table, Extensible, and Updated. The first row is highlighted.

Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Incident WorkFlow	u_incident_workflow	(empty)	false	2026-09-29 07:12:01
Inclusion Endpoint	cmdb_ci_endpoint_inclusion	Endpoint	true	2026-06-12 14:06:28
Incoming App Versions	sn_um_incoming_app_version	(empty)	false	2026-06-12 15:08:39
Incoming Store Apps	sn_um_incoming_app	(empty)	false	2026-06-12 15:08:37
Incompatible Guarded Scripts	sys_gs_incompatible_scripts	(empty)	false	2026-06-12 13:57:31
Incremental Offline Deny List	sys_sg_incremental_offline_exclusion_list	(empty)	false	2026-06-12 15:01:24
Incremental offline result	sys_sg_incremental_result	(empty)	false	2026-06-12 15:01:24
Index Explanation	sys_index_explain	MySQL Explain Plan Step	false	2026-06-12 14:04:05

Name ↑
Owner
Date modified
File size

Columns
Controls
Application Access

Extensible ☐
Live feed ☐
Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format
Prefix
Number
Number of digits
Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.
Create access controls ☒
*: User role

Delete
Update
Delete All Records

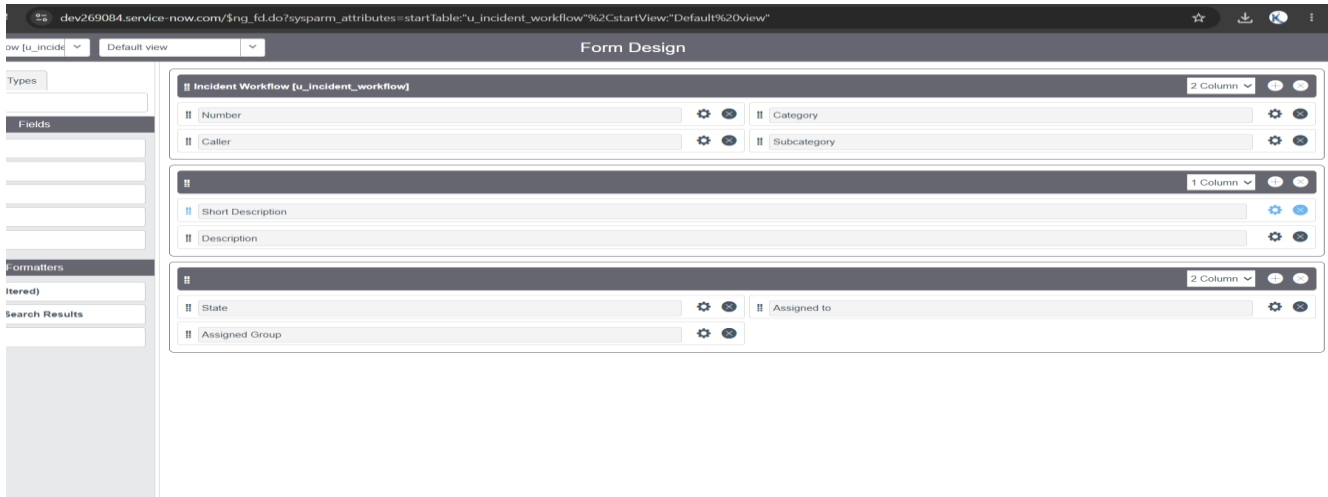
Activity 2.2: Field Creation and Data Type Configuration

The objective of this milestone is to create all required fields in the table with the correct data types, choices and dependencies, so that data is captured in a structured and consistent way.

Field Label	Type	Reference / Choice
Number	Auto Number	-
Caller	Reference	sys_user
Category	Choice	[Network, Hardware, Access, Performance]
Subcategory	Choice	[Wi-Fi, Projector, Forgot Password, Slow Computer]
Short Description	String	-
Description	String	-
State	Choice	[New, In progress, On hold, Resolved, Closed]
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- After creating the fields with the appropriate types, add the choices from the Form Designer to all choice list fields: **Category, Subcategory and State**.

- Click the **Save** button on the top right corner, then open the form and reload it.



Activity 2.3: Implementing Dependency Between Category and Subcategory Choice Fields

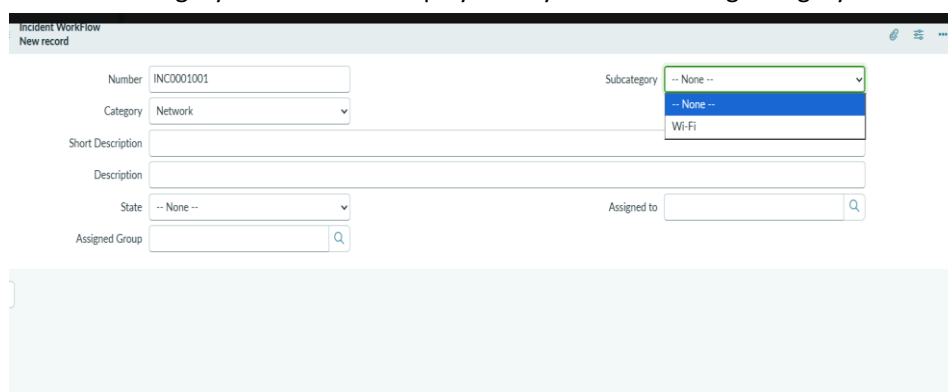
The objective is to make the available Subcategory options change dynamically based on the selected Category.

Navigation steps:

- Open the created table form and right-click the **Subcategory** field.
- Click **Configure Dictionary**.
- Under Related Links click **Advanced view**.
- Open the **Dependent Field** section, enable **Use dependent field**, and select **Category** in Dependent on field. Click **Save**.
- Scroll to the Choices section and, for each Subcategory choice, set the Dependent value to the backend name of its Category:

Subcategory	Dependent value (Category)
Wi-Fi	Network
Projector	Hardware
Forgot Password	Access
Slow Computer	Performance

- Click **Save**. Each Subcategory choice is now displayed only for its matching Category.



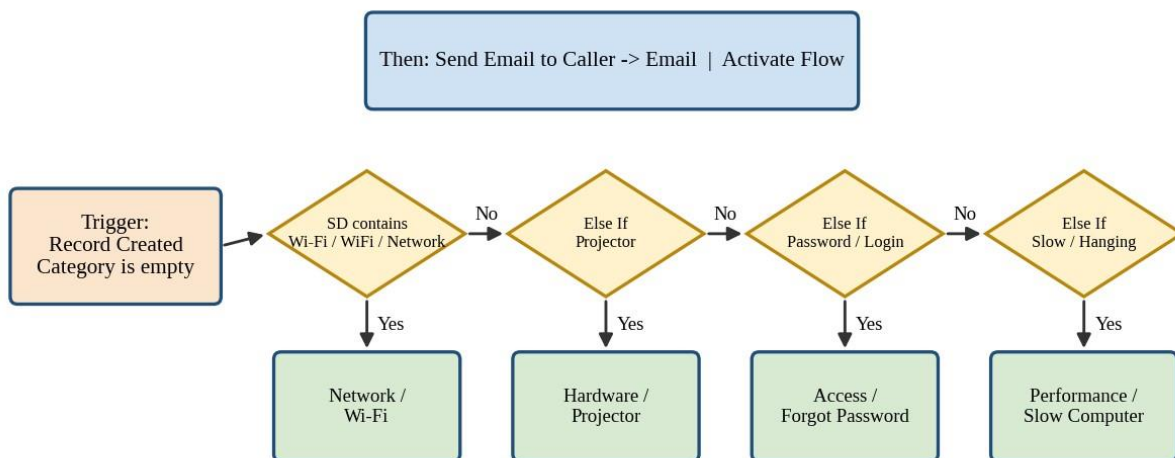
Milestone 3: Automation using Flow Designer & Email Notification

When a student creates a new IT support ticket, they only need to enter the Caller Name and Short Description. They do not need to select the Category or Subcategory. Once the ticket is submitted, the Flow Designer automation is triggered and checks the Short Description for predefined keywords such as WiFi, Projector, Password or Slow. Based on the keyword found, the flow assigns the Category and Subcategory:

- Short description contains *WiFi* (or Wi-Fi / Network): Category = **Network**, Subcategory = **Wi-Fi**.
- Contains *Projector*: Category = **Hardware**, Subcategory = **Projector**.
- Contains *Password* or *Login*: Category = **Access**, Subcategory = **Forgot Password**.
- Contains *Slow* or *Hanging*: Category = **Performance**, Subcategory = **Slow Computer**.

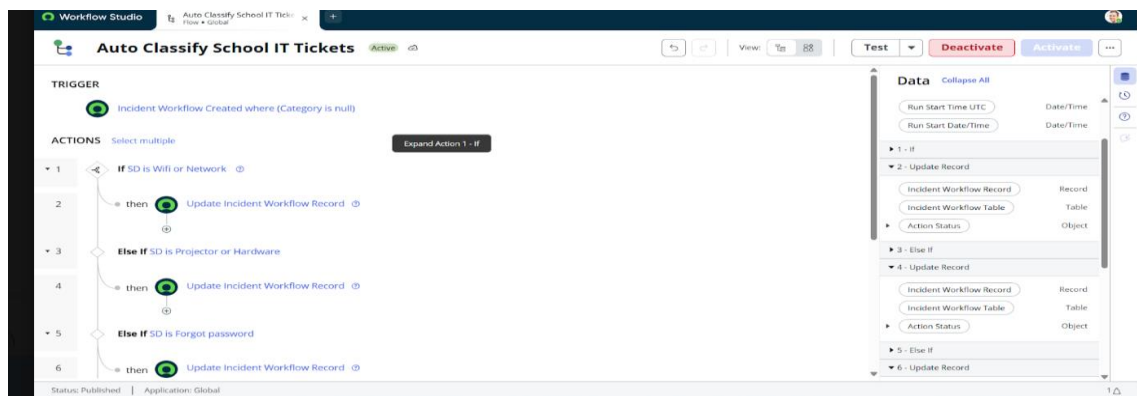
This automation gives consistent classification, removes human error and greatly reduces the manual effort of IT staff.

It is implemented fully with Flow Designer, so it is easy to maintain, scalable and completely no-code.



Activity 3.1: Auto Classification using Flow Designer

- Go to All menus and search for **Flow Designer**. Under Process Automation open **Flow Designer**.
- On the top right corner click **New** and select **Flow**.
- Name: **Auto Classify School IT Tickets**. Application: **Global**.
- Click the **Build Flow** button.

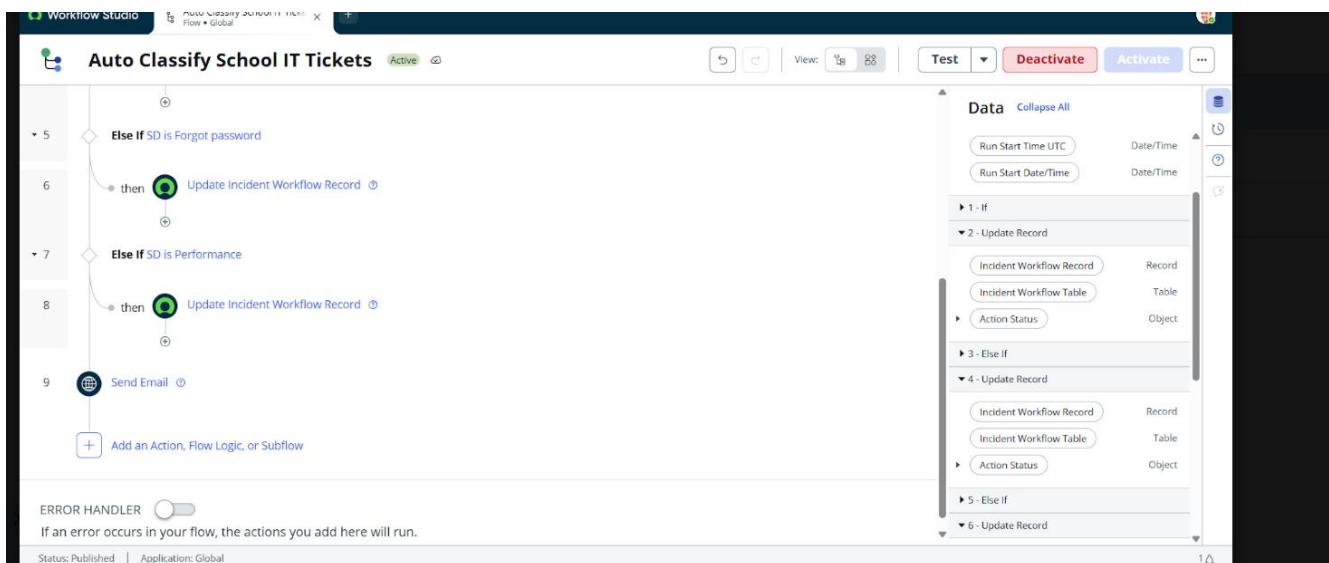


Activity 3.2: Trigger Configuration

- Trigger: **Record Created**. Table: **Incident Workflow**. Condition: **Category is Empty**. Click **Done**.
- Under Actions select the **Flow Logic** option.

Activity 3.3: Wi-Fi Issue Classification

- Select the **If** condition. Condition label: **SD is Wifi or Network** (any clear label can be used).
- Condition 1: using the data pill picker select **Short description** as the field, operator **Contains**, value **Wi-Fi**. (Add the same condition with value **WiFi** so both spellings match.)
- Condition 2: field **Short description**, operator **Contains**, value **Network**. Click **Done**.
- Click Action and select **Update Record**. Drag and drop the record from the data pills on the right; Table updates to Incident Workflow automatically.
- Set **Category = Network** and **Subcategory = Wi-Fi**. Click **Done**.

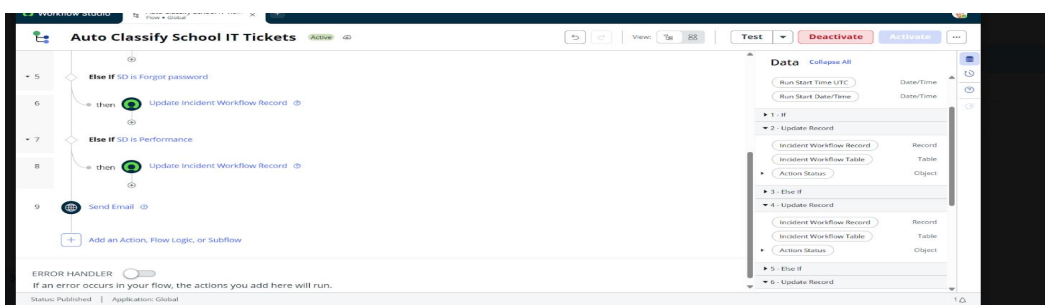


Activity 3.4: Projector Issue Classification

- Select the **Else If** condition. Label: **SD is Projector or Hardware**.
- Condition: **Short description Contains Projector**. Click **Done**.
- Action: **Update Record**. Set **Category = Hardware** and **Subcategory = Projector**. Click **Done**.

Activity 3.5: Password Issue Classification

- Select another **Else If** condition. Label: **SD is Forgot password**.
- Condition: **Short description contains Forgot password** (also Password / Login). Click **Done**.
- Action: **Update Record**. Set **Category = Access** and **Subcategory = Forgot Password**. Click **Done**.

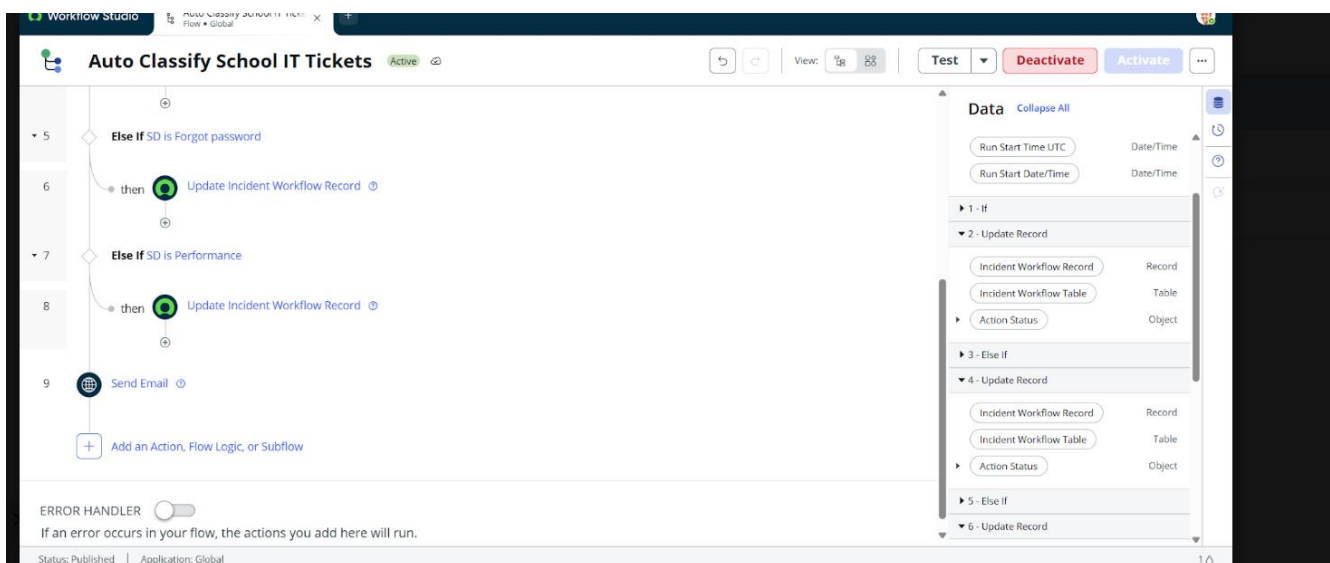


Activity 3.6: Slow Computer Classification

- Select another **Else If** condition. Label: **SD is Performance**.
- Condition: **Short description contains Slow computer** (also Slow / Hanging). Click **Done**.
- Action: **Update Record**. Set **Category** = **Performance** and **Subcategory** = **Slow Computer**. Click **Done**.

Activity 3.7: Email Action Setup

- After classification, an email notification is sent using Flow Designer. Add the action **Send Email**.
- To: **Caller > Email**. Subject: **Your Request for the issue has been submitted**. Body: **Ticket confirmation message**. Click **Done**.
- Click the **Activate** button from the top right corner.



Milestone 4: Testing, Validation & Security

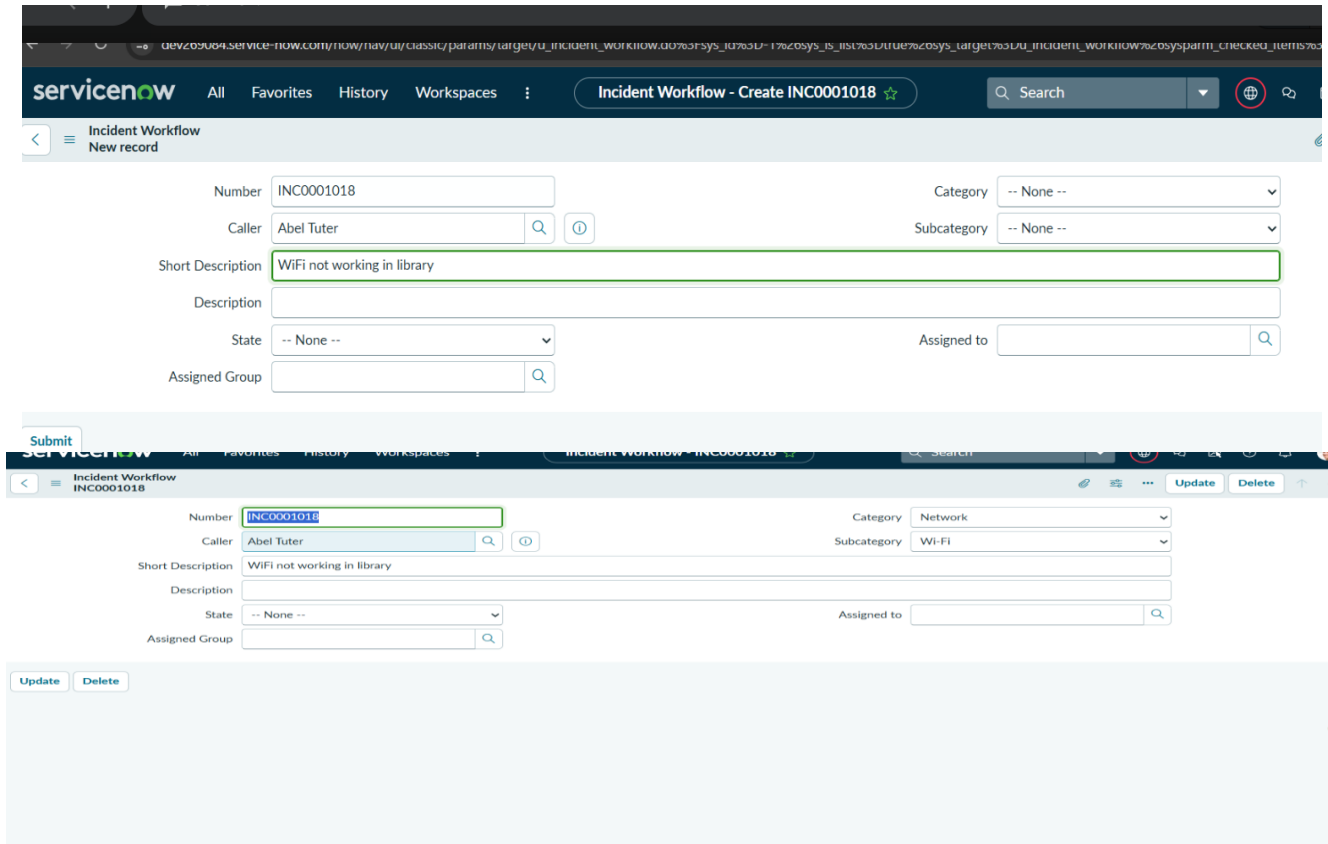
The objective of this milestone is to ensure the solution works as expected, meets all functional requirements, and follows basic security and access control practices. It validates the end-to-end automation, confirms data accuracy, and ensures only authorized users can access or modify ticket data.

Data validation checks:

- Mandatory fields are captured correctly.
- Auto Number is generated without duplication.
- Category and Subcategory values are stored accurately.
- Reference fields resolve correctly to the user and group tables.

Activity 4.1: Test Scenario 1

- Enter one Caller Name. Short Description: **WiFi not working in library**.
- Save the form and reload it. It fills automatically: Category = **Network**, Subcategory = **Wi-Fi**.
- An email is sent to the caller's email id.



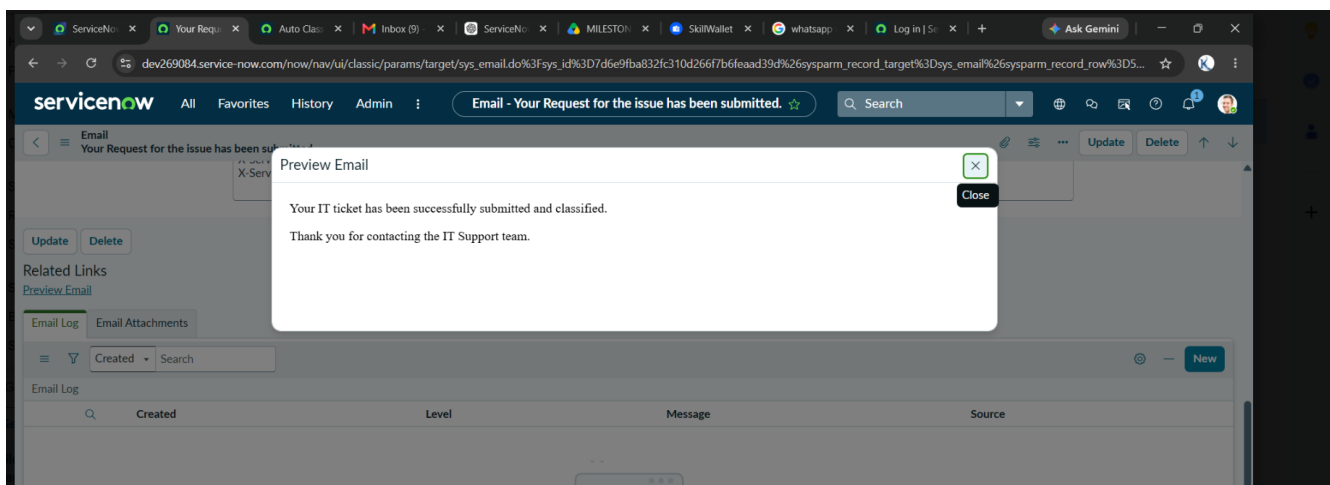
The screenshot shows the ServiceNow 'Incident Workflow - Create' form for incident INC0001018. The form is in the 'Create' state. The following fields are visible:

- Number:** INC0001018
- Caller:** Abel Tuter
- Short Description:** WiFi not working in library
- State:** -- None --
- Assigned Group:** (empty)
- Category:** -- None --
- Subcategory:** -- None --

Buttons for 'Submit', 'Update', and 'Delete' are visible at the bottom of the form.

Activity 4.2: Check Email Notification

- Open All menus and search for **Emails**. Under System Logs select **Emails**.
- Search with the given subject name, open that record and under Related Links click **Preview mail**.



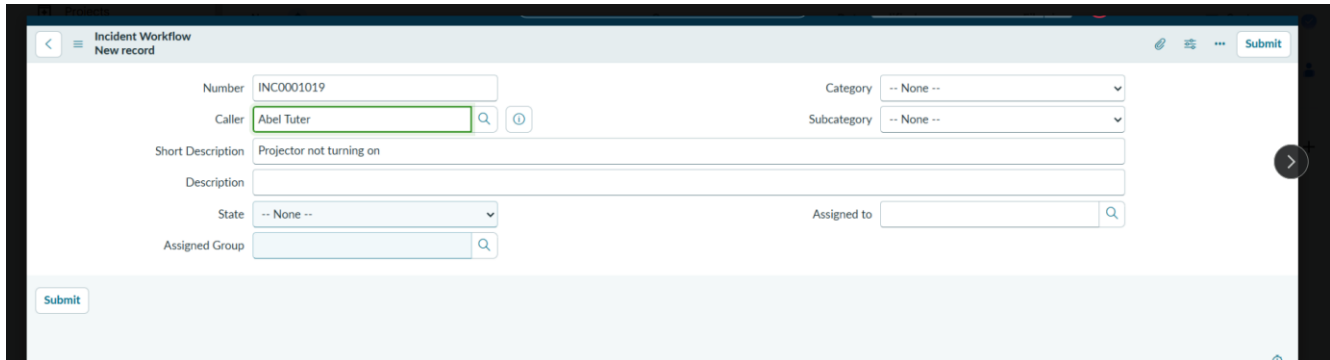
The screenshot shows the ServiceNow 'Email' record page for the subject 'Your Request for the issue has been submitted'. A 'Preview Email' modal is open, displaying the email content:

Your IT ticket has been successfully submitted and classified.
Thank you for contacting the IT Support team.

The background shows the 'Email Log' table with columns: Created, Level, Message, Source.

Activity 4.3: Test Scenario 2

- Enter one Caller Name. Short Description: **Projector not turning on.**
- Save the form and reload it. It fills automatically: Category = **Hardware**, Subcategory = **Projector**.
- An email is sent to the caller's mail id.



The screenshot shows the 'Incident Workflow New record' form. The fields are filled as follows:

- Number: INC0001019
- Caller: Abel Tuter
- Short Description: Projector not turning on
- Description: (empty)
- State: -- None --
- Assigned Group: (empty)
- Category: -- None --
- Subcategory: -- None --
- Assigned to: (empty)

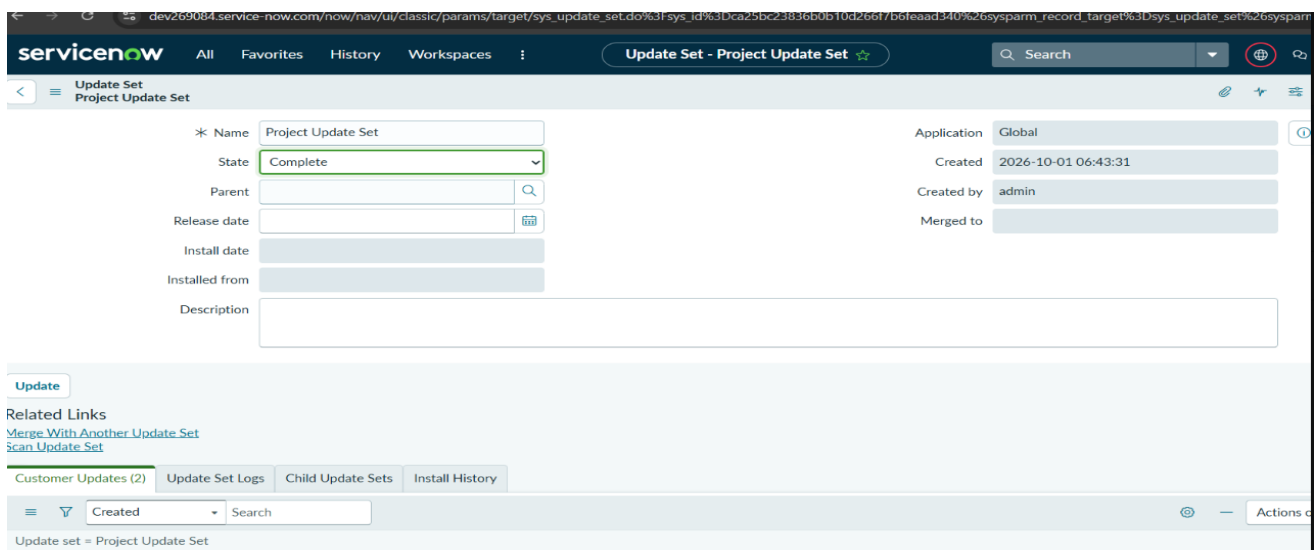
A 'Submit' button is located at the bottom left of the form.

Milestone 5: Deployment & Final Documentation

The objective of this milestone is to deploy the solution to the target environment and finalize all project documentation, so the solution is stable, reusable and ready for demonstration or production use.

Activity 5.1: Make Update Set to Complete State

- Go to All menus and search for **Local Update Sets**. Under System Update Sets select **Local Update Sets**.
- Open the **Project Update Set**. Change the State from **In progress** to **Complete** and **Save**.
- Under Related Links click **Export to XML**. The update set is downloaded to your system as an XML file.
- The downloaded file can be shared through WhatsApp or mail and loaded into another instance in the same way.



The screenshot shows the 'Update Set - Project Update Set' form in ServiceNow. The fields are filled as follows:

- Name: Project Update Set
- State: Complete
- Parent: (empty)
- Release date: (empty)
- Install date: (empty)
- Installed from: (empty)
- Description: (empty)
- Application: Global
- Created: 2026-10-01 06:43:31
- Created by: admin
- Merged to: (empty)

Below the form, there is an 'Update' button and a 'Related Links' section with links: [Merge With Another Update Set](#) and [Scan Update Set](#).

At the bottom, there is a tabbed interface with 'Customer Updates (2)', 'Update Set Logs', 'Child Update Sets', and 'Install History'. The 'Update Set Logs' tab is selected, showing a table with columns: Created, Search, and Actions.

Conclusion:

The Auto Ticket Classification using Flow Designer project delivers a complete, end-to-end automation for a school IT helpdesk. The main aim of reducing manual effort in ticket categorisation and improving response efficiency has been achieved.

Through structured requirement analysis, careful backend configuration and no-code automation with Flow Designer, the system now classifies IT tickets automatically from the short description entered by students. Dependent choice fields keep the data accurate, and the automatic email notification acknowledges every ticket to the caller instantly.

The solution was built in phases, which kept it clear, stable and easy to maintain. Testing and validation confirmed that it works reliably, keeps data integrity and follows access control practices. Overall, the project shows how ServiceNow can be used to build scalable, maintainable and user-friendly automation without complex scripting or machine learning. It suits small and medium environments such as schools and colleges, and can be extended in future with assignment automation, SLA tracking or Predictive Intelligence.